



Smoke Test VS Sanity Test

The Health Check vs. The Deep Dive

 Tags: QA, Testing, Release Management

 Reading Time: 3 min

TL;DR (Too Long; Didn't Read)

Smoke Testing verifies the critical functions work (is the build stable?).

Sanity Testing verifies the new fixes work (did we solve the problem?).

The Core Difference

Many testers treat these as synonyms. They are not. The difference lies in the **goal** and the **depth**.

Smoke Testing (The General Health Check)

Goal: To make sure the app isn't completely broken.

- **Done First:** Before any other testing.
- **Checks Critical Features:** Just the basics (e.g., Can I log in? Does the homepage load?).
- **The Rule:** If the smoke test fails, we stop working and send it back to the developers.

Sanity Testing (The Logic Check)

Goal: To verify specific changes.

- Performed after receiving a software build with minor changes or bug fixes.
 - "Narrow and Deep" - focuses only on the modified components.
 - Checks that the specific bug was fixed and didn't break immediate logic.
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The Car Mechanic Analogy

Think of taking your car to the garage because the air conditioner was broken.

The Smoke Test:

The mechanic turns the key. Does the engine start? Do the wheels turn? Does the car drive?

If the car doesn't start, there is no point checking the AC.

The Sanity Test:

The mechanic turns on the AC to max cool. Does cold air come out?

Verifying the specific fix was successful.

The Comparison Matrix

Scenario 1: New Build Arrives

- Type:  Smoke Test
- Action: Check login, homepage loads, and database connection.
- Why: We need to know if the application is stable enough to test. If these fail, the build is rejected.

Scenario 2: Developer Fixes "Cart Bug"

- Type: 🧠 Sanity Test
- Action: Add item to cart, change quantity, remove item.
- Why: We are verifying the specific functionality that was broken to ensure the fix works.

Scenario 3: The Order of Operations

- Order: Smoke First ➡️ Sanity Second ➡️ Full Regression
- Why: You cannot verify a specific fix (Sanity) if the system is crashing on launch (Smoke).